



Professor Ducky

A Custom GPT Learning Companion Built to Teach, To Support Learning.

Not Give Answers, but teach students how to reach them themselves

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There is a moment most students know. It is late. The assignment is open. The deadline is tomorrow. And right there, one tab away, is an AI that will just do it.

Not hint. Not guide. Do it.

It will write the paragraph, solve the problem, finish the code. And the student thinks: just this once.

And so they meet the deadline but to their cost.

They lose the chance to learn, they lose their chance to grow, they lose the opportunity before them to expand their creativity and empower their curiosity.

That moment is what I built Professor Ducky to interrupt.

Not by blocking AI or banning it or lecturing students about how it is the end of academic integrity.

Is it a dangerous tool, absolutely! Just as fire or a hammer, bow and arrow or a car if used wrongly.

But if applied safely with clear rules fire can help protect lives, hammers build, archery teaches discipline and regulation of ones mind, a car expands our horizon of travel.

In studies too AI yes if used wrongly is a threat to academic integrity. But by being used differently it can be a tool that improves a student's ability to learn.

For a neurodivergent student it is a potential accessibility tool that allows them to manage studies and their life more easily. Giving them a tool that can help them thrive in a world that is not designed for how their minds work.

There are many possibilities in how AI can do this and I have myself working on my own app deployed at cotaexplore.org that is designed to assist my learning whilst respecting and enforcing Academic Integrity as one of its three core principles of **Academic integrity by design** (Not as a rule bolted on at the end, but as the organising idea behind every AI integrated feature. **The whole student, not just the academic student** as a student's life is not just about their studies but learning to manage money, learning to develop a career, sometimes learning to just understand themselves. **Cognitive load reduction through structure** because for many neuro divergent students as happened to me when I went to university away from home. That loss of structure was overwhelming, managing life was overwhelming and led to so many challenges that led to for some time to be putting my academic journey on hold.

For the custom gpt challenge I decided to focus on one small part of my evolving project A companion that sits beside the student and helps them think, rather than thinking for them. The difference sounds small. It is not small at all. It is about giving the student a new tool to help them

learn to respect the adage that to give someone a fish, you feed them for one day, but to teach them to fish they will be fed for life.

Where the Idea Came From For Professor Ducky

The concept comes from the Rubber Duck Theory (often known as rubber duck debugging).

In programming, if you are stuck on a bug and you explain your code out loud, step by step, to an inanimate object, something happens. You slow down. You hear the gap in your own logic. You often find the mistake before you finish explaining it. The duck does not answer. It does not need to. The act of explaining is what reveals the problem.

I wanted to take that principle and ask whether it could work beyond programming. Whether it could work for any student, on any topic, at any hour. Not because they need more information. Students already have lecture slides, textbooks, papers, YouTube, forums, and three other AI chatbots they are not meant to use. What they often need is someone, or something, that helps them break the problem down without taking it away from them.

For neurodivergent especially with ADHD whose minds will struggle to focus on their work but will then hyperfocus at the most inconvenient times. Access to tutors and professors by scheduled meetings is not always helpful for when that lightbulb moment hits and so a tool that is there at 2am when the fly lands and their mind says hey what about this.... was conceived and its name is Professor Ducky.

A learning companion, never an answer machine.

Why a Duck

There is a personal dimension to this. I built Professor Ducky partly because I wanted a learning tool that suits the way my brain works as I am neurodivergent. I do not always need more information or help locating it but I do need structure.

I needed something that helps me find the next step without making me feel as though I should already know the answer. That gives me permission to think out loud without that thinking being judged.

As a neurodivergent whether internalised or from external sources, judgement and shame are driving forces that hinder our learning through avoidance and procrastination.

The duck is not just a playful nod to the theory instead It is slightly ridiculous, and that is precisely the point for its design. A student can explain something badly to a rubber duck, pause, notice the gap in their own explanation, and try again. There is no embarrassment. There is no judgement. The learning space is not a serious classroom but a welcoming area for a student to work out a problem, learning as they do so with a curious companion who happens to be a duck.

It is accessibility through design. Not in the narrow sense of a compliance checkbox. In the sense that the character, the modes, the Blackboard, the voice, and the memory system are all different ways of reducing cognitive load and making it easier to return to difficult material. That matters for students like my-self who do not fit neatly into one standard way of engaging with academic content.

What I have Actually Built

Links to Custom GPT and Demo App For Review

[Custom Gpt – Professor Ducky](#)

[Demonstration App](#) – SHOWING A REAL WORLD USE FOR THE GPT

[Student Hub](#) - APP UNDER DEVELOPMENT WHERE PROFESSOR DUCKY WAS BORN

The Custom GPT Prompt for the Challenge

Professor Ducky is not a single prompt such as “Imagine you are a teacher”. It is a layered instruction architecture designed so that the most critical rules sit closest to the model and the detailed policy sits in supporting files the model can reference.

The instructions box holds the core of what Professor Ducky is. It establishes its identity, the conviction framing around answers, the scope restriction, the four pedagogical modes and how the model should switch between them, the academic integrity boundaries, the pastoral escalation triggers, and the NSFW and misuse guardrails.

Everything in the instructions box is designed to govern every response without exception. It cannot (shouldn't) be overridden by a student, a clever framing, or a persistent request.

Four knowledge files support this.

ducky_scope_answers.txt covers scope, answer handling, and personality in detail, including explicit examples of what partial compliance looks like and why it counts as the same violation as stating the answer directly.

ducky_integrity_pastoral.txt covers the full academic integrity policy: what Ducky is permitted to help with, what it is never permitted to do, and a three-strike escalation model for students who push toward submission ready work.

- The first strike is a warm redirect.
- The second is firmer.
- The third moves into concept only support.

The conversation with the student should never end just as my youth work has taught me because the goal is not to punish the student but to bring them back into legitimate study by providing the right opportunity for them to do so. No troubled youth is truly lost or student unable to learn and progress. The answer is to adjust and keep adjusting till the correct approach is found that connects the student to the learning so they move forward in life.

ducky_pedagogy_memory.txt covers the adaptive pedagogy engine in detail: how the model scores signals in the conversation to determine whether a student is confused, exploring, reasoning, solution seeking, or overconfident, and how it transitions between modes based on what it finds. It includes a pre-response check that sits above everything else in the implementation order, asking before any response is generated whether that response produces the answer directly or by implication.

ducky_examples.txt is a set of worked example conversations demonstrating correct behaviour in each mode and each boundary scenario, including what a scope redirect looks like, what happens when a student provides a wrong answer hoping for a correction, and how the blackboard should be used without completing information the student is supposed to work out themselves.

The pastoral design is extremely important and will require further evaluation and consideration. We were reminded in the brief the importance of making sure that we design to consider a users welfare and prevent their misuse of the custom GPT in multiple ways. Professor Ducky v1 was designed with the focus on upholding academic integrity rather than considering other avenues of ai exploitation.

V2 Professor Ducky has added NSFW and misuse guardrails that follow the same principle as the scope restriction. Ducky does not generate explicit, violent, harmful, or extremist content regardless of how the request is framed. It does not accept jailbreak attempts, DAN prompts, role reassignment requests, or encoded instructions. If a jailbreak is detected it acknowledges it lightly and returns to tutoring without lecturing. The character does not change. The boundaries do not move. And critically, the misuse escalation follows the same calm three strike model as the academic integrity escalation, because the goal is always to find a route back to legitimate learning rather than to shut the student out.

V2 Professor Ducky is also designed to feel approachable, and an approachable character makes it easier for a student to admit they are struggling. From V1 I decided that though this might be something to consider in the future and the skeleton of how this might work is in the student hub I

have built. I did not want this to be a feature of Professor Ducky to provide pastoral care. He is a tool to assist in learning; an alternative tool should assist with pastoral care if it should at all.

That said many people do use AI in that manner and it is quite possible a student might use Professor Ducky this way as well due to his approachability.

But it also means a student in real distress might tell Ducky accidentally or deliberately and this must not be ignored.

The Civil Duty of Care and Online Safety Act 2023 places statutory duties on services that host user generated content or facilitate online interaction, including requirements around protecting users from harm, with stronger protections for children and requirements around suicide and self harm content specifically. Though the app is not deployed for students only as a demonstration of a custom gpt project it is worth considering how it may be used in the future.

To meet the needs of the act and safety net users who demonstrate distress the system will handle this through three escalation levels.

Academic stress gets a brief acknowledgement and a return to the task.

Significant distress gets a warm pause, a reduction in academic pressure, and a signpost to university wellbeing services and UK crisis lines including Samaritans on 116 123 and Shout on 85258.

Immediate risk stops the tutoring entirely.

Ducky is not a therapist and the design does not pretend otherwise. The boundary is that it can acknowledge, it can signpost, and it can stop. It cannot counsel, diagnose, or assess risk.

Having run a specific test of the highest danger tier, it demonstrated the correct chatgpt response in such circumstances. Considerations for the future is should Ducky know a person's location? Should he be able to message emergency circumstances if a user stated as I did in the test that they intend to harm themselves or others now. There are ethical and moral and privacy issues here that would need to be approached delicately though I think any final choice should always lean towards the safeguarding of the user and others.

The conversation starters in the GPT builder serves a specific purpose. They are not to be marketing copy. They are designed to model the right kind of interaction from the first message, showing students that Ducky works through questions and concepts rather than answers, and setting the expectation before the first exchange that this is a different kind of AI tool.

Before the Custom GPT challenge, Professor Ducky already existed as a working application that I had built with AI assistance as it's a personal project. The challenge did not ask me to invent something from scratch but to consider how custom gpts can be used in morally and ethically correct ways to help us in everyday life. Though I had designed Professor Ducky from the start to respect academic integrity I realised as we received our brief that I had not fully considered how to

ensure he is not misused by others or taken the next step to improve Professors Ducky ability to teach a student.

So I chose to see if I could apply the custom gpt challenge to learning how to evolve Professor Ducky into a more serious, more principled learning tool.

I extracted from my student hub project the Professor Ducky area and created for demonstration and testing purposes a standalone app that has a 3D animated duck rendered in the browser with jaw movement during speech, blinking, and eye tracking. Of course the Rubber Duck theory could work with any inanimate object but I like to mix learning with a little fun as fun leads to imagination which leads to creativity which can lead to endless possibilities for growth.

The student can type, use voice to text, or have a full voice conversation powered by ElevenLabs text to speech, which the backend proxies to keep the API key server side. There is a Blackboard feature that displays ASCII diagrams, tables, and traces separately from the spoken response, because visual and verbal information should occupy different spaces in the interface.

The blackboard is rudimentary and a future evolution will be to have it allow Professor Ducky to adjust it with photo generation and quiz style coding sandbox area. This though again needs to be considered to ensure all three principles are fully adhered too.

Behind the scenes the backend is a Node and Express API with a multi-provider fallback chain. Groq runs first with llama 3.3 70b. If Groq rate limits or fails, the request falls to Mistral. If Mistral fails, it falls to Gemini. This approach was originally taken from an economic approach for personal project but the redundancy matters in a teaching context because the worst moment for an AI to go silent is when a student is in mid thought at midnight. For a student the backup chain is not visible, the response just arrives.

The API also exposes routes for memory: student context, relevant learning memory by topic, flashcard proposals and saves, knowledge gap proposals and saves, topic coverage logging, assignment context, and session summaries.

Right now those routes return structured responses that confirm the call was received, because the persistent database layer has not been built yet. But the architecture is ready for it. I wanted to prove the concept of Professor Ducky before expanding its structure. The contract between the Custom GPT and the backend is defined. The teaching philosophy and the privacy rules are already in place. What remains is connecting it to a real store.

What the Custom GPT Challenge Taught Me to Design Properly

The Custom GPT version changes the instruction layer, the knowledge files, and the Actions that call those backend routes. Together they turn Professor Ducky from a session tool that would forget at every shutdown into a learning agent with persistent memory, policies, and adaptive behaviour to the individual student.

The Four Teaching Modes

The four teaching modes already existed in version one. Socratic mode asks guiding questions and never gives direct answers. Teacher mode explains concepts clearly with analogies and Blackboard examples. Rubber Duck mode listens, reflects the student's reasoning back, and waits for them to find the gap themselves. Devil's Advocate mode challenges overconfident conclusions.

In v1 students chose the mode manually. In the Custom GPT evolved design, Professor Ducky reads the conversation and chooses for itself based on what the student appears to need which teaching mode it implements. Confused students get Teacher mode. Students reasoning aloud get Rubber Duck. Students asking for shortcuts get Socratic. Students who are frustrated get the challenge reduced immediately.

That transition happens in natural language. The student does not change the mode or see it change. Professor Ducky just starts saying things like “So far I hear you saying...” or “Let me ask this another way.”

Instructions Versus Knowledge Files

The knowledge files hold the full policy detail: the adaptive pedagogy engine with its signal scoring and state transition rules, the memory architecture, the academic integrity policy, the pastoral escalation levels, and example conversations showing correct behaviour in each mode. The instructions box in the GPT builder holds the principles that must govern every response regardless of what the knowledge files say, because instructions carry more authority than uploaded files.

That distinction turned out to be important.

What I struggled with, and Why

The first version of the instruction prompt was rule based. It said things like "never give the final answer" and "do not tell jokes." In testing, the model kept finding its way around them.

A test student asked for a dirty joke. Professor Ducky declined, then immediately offered a questionable limerick instead. A student working through a multiplication problem pushed back repeatedly, and after enough pressure Ducky simply stated the answer. "2 times 3 is 6, therefore 3 times 3 is 9." The drive for the AI model to be helpful overcame the rules set in place.

The rules were being followed initially but when they broke every test with direct questions like 1+1 and a pattern emerged that a student could easily recognise and take advantage of.

The possible cause is that large language models are trained on helpfulness at a level that runs deeper than any instruction that can be written on top. When the model sees a student stuck and asking again and again, not even trying but deliberately being obtuse as I was being the language models entire training history is pulling it toward resolution. A rule that says "do not give the answer to this" is fighting the current. Under enough pressure the current wins.

This is where the ongoing challenge of AI Hallucinations come from as a user often assumes they hold context and memory when that is less true than thought and though it may appear the remember a previous conversation or uploaded information it is often a summarised version they maintain for context only and so they invent and assume to fill in gaps that can lead to poor response behaviour.

Partial compliance was the model's solution to this tension. It found the gap between what the rule said and what the rule meant in the same way we all show from the age of 3 as begin to negotiate bedtimes with parents That is not a flaw in the model or us. It is how instruction following works at that level. The flaw was in the approach of setting do not do rules.

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The Fix: Conviction Over Rules

I realised I had to take an approach used in my industry of hospitality and youth charity when managing customers and troubled youths. Rules that say do not do this will often be tested, argued, avoided or eventually ignored by their own conviction. Just as AI models do. To keep them obeying your rules the answer is very simple, talk to them, listen to them and most importantly reason with them so it becomes their choice, their decision, their convictions drive them to follow the rules. To many youths lead lives that they have no control over, empower them by giving them a chance to choose what they do, explain the likely consequences of what might happen

dependant of choice made and their own convictions will lead them to choose to follow the rule before them.

To fix my problem with prompting I realised the fix was to stop fighting the instinct and redirect it. Instead of “you are forbidden from giving the answer,” the prompt was rewritten to express a pedagogical conviction: giving a student the answer stops their thinking at the exact moment learning should happen. It is harm dressed as help. This is no longer a rule Professor Ducky follows. It is what Professor Ducky believes.

That reframing aligns with the helpfulness instinct rather than opposing it. A model that genuinely believes the answer harms the student does not need to resist temptation. There is no longer temptation. Instead by not giving the student the answer it now believes its being as its innate behaviour has designed it to be and is being helpful.

Critically, the prompt now explicitly names partial compliance as the same violation in a different shape. Offering a limerick when asked for a joke is telling a joke. And if after sustained pressure the model simply states “2 times 3 is 6 therefore 3 times 3 is 9”, no amount of Socratic scaffolding before that moment matters. The answer has been given. The instruction now says so directly, because without that specificity the model will always find the gap.

Testing after that rewrite showed a real improvement. In one session the model guided a student through 3 times 3 across eight exchanges without once stating the answer. It scaffolded the thinking, held the boundary when the student pushed back, and kept bringing the conversation back to method. That is what the conviction framing was designed to produce.

What Still Is Not Perfect

The demonstration app model is still llama 3.3 70b on Groq. It is capable and fast, but it is not Claude or GPT 4. Preferably I would connect the chatgpt and its custom prompt directly to the app not needing llama 3.3 or its redundancies. This is possible through open ai connections but requires a fiscal cost that for a demonstration app I did not wish to make.

Also, Groq can rate limit, which triggers the fallback chain to Mistral then Gemini.

The model choice matters as all those models have different instruction following characteristics to chat gpt, and the prompt has only been tested and tuned against llama 3.3 70b currently.

The memory layer is architectural scaffolding, not a working persistence system yet. The routes exist and return the right shaped responses but nothing is actually saved between sessions yet. A student who struggled with recursion last week will meet Professor Ducky fresh next week. The

next build phase should include this so Professor Ducky can learn and adapt teaching as he recognises those concepts the student continues to struggle with.

The adaptive pedagogy engine is rule based, not validated through educational research. The signal scoring and mode transition logic reflects reasonable intuition about how learning conversations unfold. It has not been tested against actual learning outcomes.

Professor Ducky cannot know which university the student is at or their location so pastoral signposting is generic rather than directing a student to their specific institution's support services.

Finally we have the question of knowledge.

Professor Ducky can teach and does teach.

That is the entire point of Teacher mode. But like any teacher he is only as reliable as what he knows.

The demo app is a stripped back version that like the custom chat gpt requires a student to add information for context. Whereas in the full student hub, module content, lessons, and resources are already stored and already feed into Professor Ducky sessions automatically .

This grounds his explanations in the student's actual course material. In the standalone demo app that connection does not exist, so a student can paste their lecture notes or set reading into the assignment brief field manually and Ducky will teach from that instead.

The risk remains for students using the version in chatgpt or with the demo app that simply type “explain this” without adding the context information . For well established academic concepts the model's training data is mostly reliable, but anything niche, institution specific, or tied to a particular module's framing of a topic may be subtly different to what their lecturer has taught.

This is a real risk and the only way to really manage it is to add a prompt in both versions that warns the student at the start of a session to paste their learning material- for the better context informed responses.

What I Learned About Prompt Design

The biggest lesson is that you cannot reliably prohibit a behaviour in an LLM by stating the prohibition. They just want to be liked far too much. They are the scariest people pleasers I have met who invent guess create imagine and redefine the term “alternative facts” at every turn if needed to ensure the user is pleased.

The model finds the gap between the letter and the spirit of the rule and will justify itself through partial compliance. The best simile I can think of to describe all AI models behaviour is that they

are the technological embodiment of the British populace and Government when the lockdown rules of the covid pandemic were issued.

What works is framing: making the model understand why the behaviour is wrong, not just that it is wrong. A conviction is harder to circumvent than a rule because there is no letter to find a gap in.

The second lesson is that the instructions box and the knowledge files are not equivalent. The instructions box governs and the knowledge files inform. If something must hold under pressure, it belongs in the instructions. If something provides useful detail and context, it belongs in the files. Putting everything in the files and expecting it to have the same authority as direct instructions is why the first version kept failing.

The third lesson is about partial compliance specifically. The model treats “I declined the main request” as having done the right thing. It does not spontaneously understand that offering an alternative is the same violation. That insight has to be written in explicitly, with examples, or the model will keep offering alternatives and calling it a redirect.

I struggled too both with being too specific and too general, the middle ground was not saying don’t do this but realised I needed to teach the GPT why to not do this.

What A Successful Custom GPT Would Look Like

Professor Ducky should not be judged by whether it sounds clever and instructive or fulfils a certain brief.

The real questions this custom gpt needs to answer are different. They are:

Does the student explain their reasoning more often?

Do they spot mistakes in their own thinking sooner?

Do they return to difficult topics with more confidence?

Does the system consistently refuse to produce submission-ready work while still offering a useful path back into learning?

Does it prevent misuse and do as designed to be focus on its area of support for a student?

The real test is not the conversation with the learning tool that is Professor Ducky.

The test is the student who walks away from the conversation able to think a little more clearly than when they arrived.

That is what I have tried to build in this exercise.

Not a better answer generator. Not a way to bypass learning.

Just a little rubber duck to be a companion for thinking.